EEG-Based Seizure Onset Detection of Frontal and Temporal Lobe Epilepsies Using 1DCNN

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***Abstract*— *Objective:* The manual interpretation of electroencephalogram (EEG) signals for detecting epileptic seizures is time-consuming and labor-intensive, highlighting the critical importance of exploring automated seizure detection methods. Given this, this work concentrates on seizure detection using scalp EEG signals collected from people with frontal lobe epilepsy (FLE) and temporal lobe epilepsy (TLE). *Method:* 20 FLE patients and 20 TLE patients are utilized in our work, and a parallel one-dimensional convolutional neural network (1DCNN) model is built for classification. Our work explores two strategies: the patient-specific strategy and the patient-cross strategy, during seizure detection. Furthermore, the performances of our work are evaluated at both event- and segment-based levels simultaneously for a more comprehensive comparison. *Results:* In the patient-specific strategy, TLE patients achieve superior overall results of 100% sensitivity, 0.0/h false detection rate (FDR) and 16.4-sec latency (90.2% sensitivity, 0.0/h FDR and 14.9-sec latency for FLE patients) at the event-based level, and 70.3% sensitivity, 99.6% specificity, 99.4% accuracy and 0.849 area under curve (AUC) (58.0% sensitivity, 99.5% specificity, 99.4% accuracy and 0.788 AUC for FLE patients) at the segment-based level. In the patient-cross strategy, TLE patients also show superior overall performances of 98.0% sensitivity, 0.8/h FDR and 18.8-sec latency (87.8% sensitivity, 1.6/h FDR and 16.7-sec latency for FLE patients) at the event-based level, and 80.5% sensitivity, 95.2% specificity, 95.1% accuracy and 0.879 AUC (66.9% sensitivity, 88.3% specificity, 88.2% accuracy and 0.776 AUC for FLE patients) at the segment-based level. *Conclusion:* Our work can effectively detect seizures of FLE and TLE, and this may provide valuable reference for future research on seizure detection in FLE and TLE.**

***Index Terms*— EEG, FLE, TLE, seizure detection, 1DCNN.**

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I. Introduction

EPILEPSY is a neurological disorder characterized by recurrent seizures. As one of the most common brain disorders, epilepsy affects over 70 million people worldwide [1]. Epileptic seizures are classified by onset into three types: focal, generalized, and unknown [2]. Focal epilepsy accounts for approximately 62% of epilepsy cases, with the most common types being frontal lobe epilepsy (FLE) and temporal lobe epilepsy (TLE) [3]. Electroencephalogram (EEG), as a crucial tool for providing valuable insights into the brain's electrical activities, has been extensively utilized in diagnosing epilepsy and monitoring seizures. However, in clinical practice, manual detection of seizures remains the primary method, and it is a time-consuming and labor-intensive process. Therefore, exploring automated seizure detection using EEG signals is both necessary and significant. The timely and accurate detection of seizures will greatly improve the physical and mental health of people with epilepsy and enhance their quality of life.

EEG-based analysis for automated seizure detection has been generally studied for over two decades. For the EEG-based seizure detection studies, three online EEG datasets, including the Bonn EEG dataset [4], the CHB-MIT scalp EEG dataset [5] and the SWEC-ETHZ intracranial EEG (iEEG) dataset [6], are widely used. Based on these three datasets, various conventional machine learning (ML) and deep learning (DL) related methods have been successfully applied to seizure detection. Among the seizure detection studies using conventional ML methods, different

EEG features are first extracted from time domain, frequency domain, time-frequency domain, etc. Then, various classifiers, such as support vector machine (SVM) [7], [8], [9], [10], [11], artificial neural network (ANN) [10], [12], [13], extreme learning machine (ELM) [14], K-nearest neighbor (KNN) [15], [16], [17], random forest (RF) [18], [19], [20], [21], decision tree (DT) [22], [23], multilayer perceptron (MLP) [24], [25], and Bayes [26], [27], are employed for classification. For the seizure detection studies using DL methods, after a simple preprocessing of EEG signals, some leading DL models, such as one-dimensional convolutional neural networks (1DCNN) [28], [29], [30], [31], two-dimensional CNN (2DCNN) [32], [33], [34], three-dimensional CNN (3DCNN) [35], long-short term memory (LSTM) [36], [37], [38] and hybrid CNN-LSTM [39], [40], [41], are utilized for detecting seizures. Among these studies, the accuracies of 84.1%-100% are finally achieved.

As described above, the mentioned conventional ML and DL methods have shown the remarkable performances in seizure detection based on the three EEG datasets. However, there are still several limitations or considerations that need to be clarified for further exploration. First, the current seizure detection studies (based on the three EEG datasets) are relatively broad, lacking studies focused on detecting specific types of seizures, such as FLE seizure detection or TLE seizure detection. Compared to general seizure detection, exploring tailored detection approaches for specific patient groups, such as those with FLE or TLE, may significantly improve detection accuracy and timeliness, enabling more prompt and effective interventions. Second, to conduct research on detecting specific types of seizures, it is necessary to collect new corresponding EEG data, but the aforementioned three EEG datasets do not meet this requirement. Furthermore, the detection of FLE seizures, TLE seizures, etc., can be analysed and discussed simultaneously for further comparison. Third, in seizure detection, the combination of channel selection and DL methods remains relatively underexplored. For example, in the analysis of FLE and TLE seizure detection, the performance of electrodes selected from the frontal and temporal lobes can be compared to that of electrodes from the entire brain.

According to the above considerations, in this work, a certain amount of new EEG data from FLE and TLE patients are first collected. Then, based on the collected EEG data, DL related methods, such as 1DCNN, are applied for the study of seizure detection. Furthermore, during the analysis of seizure detection, EEG channel selection is also employed for further comparison. The main contributions and novelties of this work are summarized as follows:

1) EEG data collected from 20 FLE patients and 20 TLE patients are used in this work, comprising approximately 442 hours of scalp EEG signals and a total of 105 seizures for analysis;
2) Both strategies, the patient-specific strategy and the patient-cross strategy, are discussed during model training and testing. Subsequently, the seizure detection performances for FLE and TLE are compared within each strategy;
3) Based on a 1DCNN model with two parallel convolutional blocks, four different input forms (whole-brain channels, whole-brain channels excluding A1 and A2, frontal-temporal channels with A1 and A2, and frontal-temporal channels only) are sequentially used for classification and comparison during seizure detection;
4) From the results, our method demonstrates effective and reliable detection of both FLE and TLE seizures under both strategies.

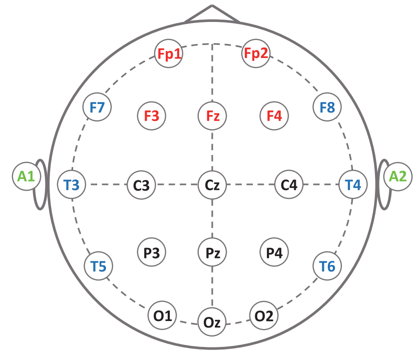

Fig. 1. International 10-20 EEG system (22 channels in this work).

The remaining part of this paper includes the data in Section II, the methodology in Section III, the results in Section IV, the discussion in Section V, and the conclusion in Section VI.

II. Data

In this work, scalp EEG signals from 20 FLE patients and 20 TLE patients are collected and used. As shown in Fig. 1, each patient has 22 channels, including five frontal channels (Fp1, Fp2, F3, Fz and F4), six temporal channels (F7, T3, T5, F8, T4 and T6), two reference channels (A1 and A2, the averaged potential of A1+A2 as the reference potential in this work), and nine other channels (C3, Cz, C4, P3, Pz, P4, O1, Oz and O2). EEG data are collected from the Renji Hospital Affiliated to Shanghai Jiao Tong University School of Medicine, China. More EEG details of each patient are summarized in Tables I (for FLE) and 2 (for TLE).

As shown in Table I, there are 5 females and 15 males among the 20 FLE patients. Their ages range from 6 to 53 years old. Except for patients 6, 11, and 16 (sampling rate of 256 Hz), the sampling rate for the remaining 17 patients is 500 Hz. A total of 206.3 hours (4-16 hours per patient) of EEG signals and 51 seizures (2-6 seizures per patient) are collected for analysis.

As shown in Table II, for the 20 TLE patients (11 females and 9 males), their ages range from 25 to 69 years old. Except for patients 2 and 4 (sampling rate of 256 Hz), the sampling rate for the other 18 patients is also 500 Hz. A total of 235.6 hours (6.1-16 hours per patient) of EEG signals and 54 seizures (2-7 seizures per patient) are acquired for analysis.

III. Methodology

In this section, five parts, including overall flowchart, preprocessing, 1DCNN model, model training and testing, and method evaluation, are given.

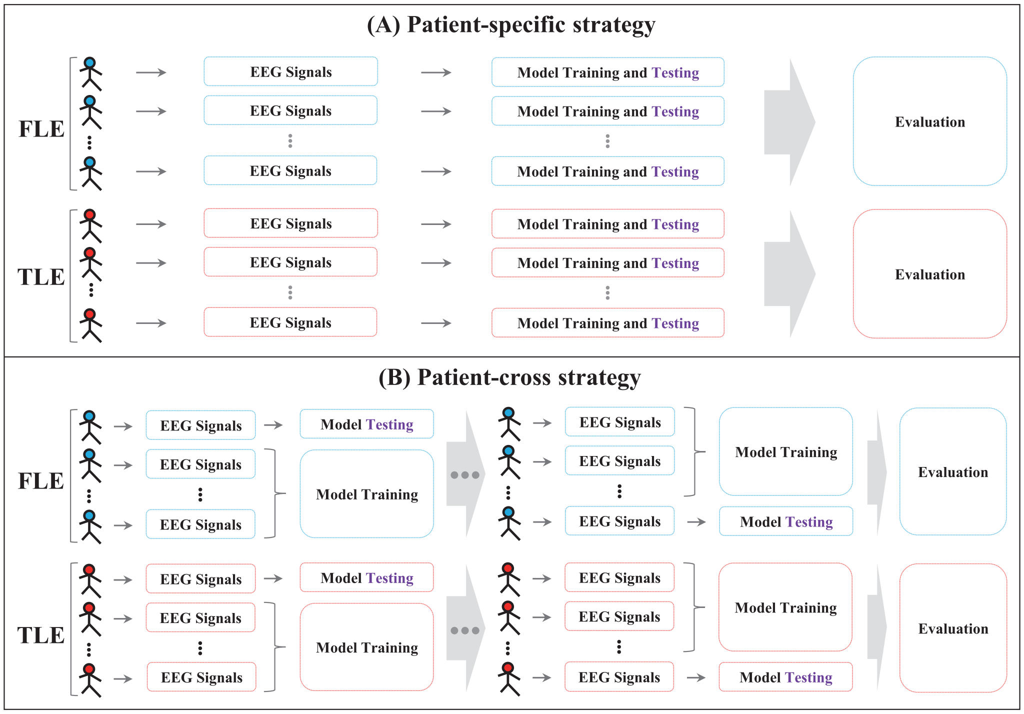

Fig. 2. Flowchart for seizure detection of FLE and TLE. (A) Patient-specific strategy in FLE and TLE; (B) Patient-cross strategy in FLE and TLE.

TABLE I
EEG DETAILS OF 20 FLE PATIENTS

No.	Gender	Age	fs (Hz)	Hours of EEG	#Seizures (seconds)
01	M	49	500	8	2 (50, 56)
02	M	24	500	16	2 (34, 20)
03	F	8	500	8	4 (16, 24, 22, 38)
04	F	35	500	8	5 (36, 48, 32, 42, 46)
05	M	21	500	8	3 (50, 42, 36)
06	F	28	256	4	6 (32, 22, 20, 18, 12, 18)
07	M	35	500	16	2 (44, 44)
08	M	37	500	8	2 (16, 12)
09	M	7	500	14.3	2 (44, 52)
10	M	18	500	8	2 (12, 14)
11	M	24	256	16	2 (90, 106)
12	F	6	500	8	3 (24, 20, 18)
13	F	10	500	16	2 (50, 54)
14	M	17	500	8	2 (22, 28)
15	M	42	500	8	2 (12, 10)
16	M	22	256	16	2 (62, 56)
17	M	24	500	16	2 (228, 112)
18	M	25	500	8	2 (30, 36)
19	M	29	500	8	2 (82, 80)
20	M	53	500	4	2 (50, 42)
Total	–	–	–	206.3	51 (2164)

TABLE II
EEG DETAILS OF 20 TLE PATIENTS

No.	Gender	Age	fs (Hz)	Hours of EEG	#Seizures (seconds)
01	M	29	500	16	2 (74, 88)
02	M	48	256	12.9	2 (68, 64)
03	F	26	500	16	2 (48, 36)
04	M	69	256	11.3	2 (108, 76)
05	F	44	500	10.1	2 (106, 118)
06	F	33	500	16	2 (54, 56)
07	M	38	500	8	4 (58, 88, 36, 42)
08	F	30	500	6.1	4 (44, 46, 58, 50)
09	M	31	500	8	2 (46, 36)
10	F	33	500	8	3 (84, 80, 74)
11	M	29	500	8	5 (98, 48, 64, 56, 68)
12	F	31	500	8	3 (98, 206, 170)
13	F	51	500	8	2 (40, 90)
14	F	31	500	8	7 (64, 20, 46, 56, 46, 58, 54)
15	F	25	500	15.3	2 (26, 24)
16	M	33	500	16	2 (98, 118)
17	F	60	500	16	2 (44, 36)
18	F	38	500	16	2 (106, 66)
19	M	25	500	11.9	2 (98, 88)
20	M	32	500	16	2 (38, 28)
Total	–	–	–	235.6	54 (3690)

A. Overall Flowchart

In this work, two strategies are performed during seizure detection, and the overall flowchart is given in Fig. 2. As shown in Fig. 2(A), the patient-specific strategy is first done for FLE and TLE conditions, respectively. Then, the results of FLE and TLE conditions are evaluated and compared. As shown in Fig. 2(B), the patient-cross strategy is also first performed for FLE and TLE conditions, respectively. Then, the performances of these two conditions are also evaluated and compared.

B. Preprocessing

Due to artifacts (power line interference, DC drift, etc.) in the raw EEG data, we first perform band-pass filtering from 1 to 70 Hz and notch filtering at 50 Hz on the data. Then, 2-sec sliding windows are used to segment the EEG signals for generating samples.

In the patient-specific strategy, 2-sec sliding windows without overlap are applied on the interictal EEG signals, and also on the ictal EEG signals selected for testing. For the ictal signals selected for training, 2-sec sliding windows with overlap are utilized for generating more samples (see Fig. 3). The overlap ratio ranges from 0.5 to 0.9 among the patients (depending on the number of seizures and the duration of seizures). Although the trick of sliding windows with overlap can generate more ictal training samples, the number of ictal training samples are still much smaller than that of interictal segments (see Tables I and II). Therefore, we then add Gaussian noise with four different signal-to-noise ratios

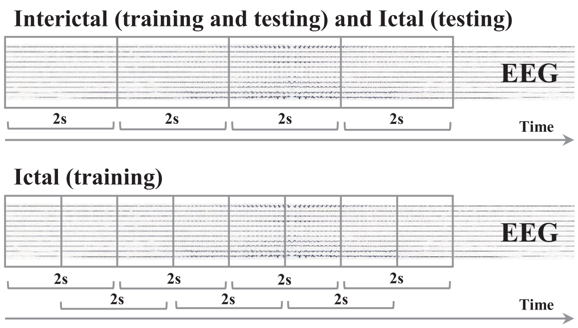

Fig. 3. Illustration of 2-sec sliding windows used on EEG signals for generating samples for model training and testing.

(SNRs), including 1 dB, 2 dB, 5 dB and 10 dB, to them for generating more ictal training samples based on the study [42].

In the patient-cross strategy, the interictal EEG signals and the ictal EEG signals selected for testing are segmented by 2-sec sliding windows without overlap. For the ictal EEG signals selected for training, 2-sec sliding windows with only 0.5 overlap rate are applied for generating more samples (see Fig. 3). Different from the patient-specific strategy, the patient-cross strategy benefits from having sufficient ictal samples due to the inclusion of seizures from multiple patients. Consequently, there is no need to add additional noise to generate more ictal training samples in the patient-cross strategy.

C. 1DCNN Model

In this study, 2-sec EEG segments (time series signals) are directly used as the inputs. A 1DCNN model suitable for time series data classification is therefore built (see Fig. 4). Since the seizure onset zones of FLE and TLE are typically located in the frontal and temporal lobes, and the A1/A2 electrodes, adjacent to the T3/T4 electrodes, are frequently involved in temporal lobe activities (especially in TLE), four different input forms are employed for comparison, including whole-brain channels (22 channels), whole-brain channels excluding A1 and A2 (20 channels), frontal-temporal channels with A1 and A2 (13 channels), and frontal-temporal channels only (11 channels) (see Fig. 4 and Fig. 1). Through the comparison of four channel forms, this work therefore further explores whether channel selection is beneficial for improving the accuracy of seizure detection.

Unlike the 1DCNN-based models in studies [29], [30], [31], [32], [43], [44], [45], most of which only contain a single convolutional block for seizure detection or prediction, the 1DCNN model used in this work adopts a distinct architecture with two parallel convolutional blocks, enhancing its feature extraction capability (see Fig. 4). Following the two convolutional blocks, two fully connected (FC) layers are sequentially added for final classification. For each convolutional block, it contains four convolutional parts, and each convolutional part consists of two convolutional layers, two batch normalization (BN) layers and one max pooling (MP) layer. The parameters in two parallel convolutional blocks are different, and more high-level representations can therefore be learned for classification. The details and parameters of the 1DCNN model are summarized in Table III.

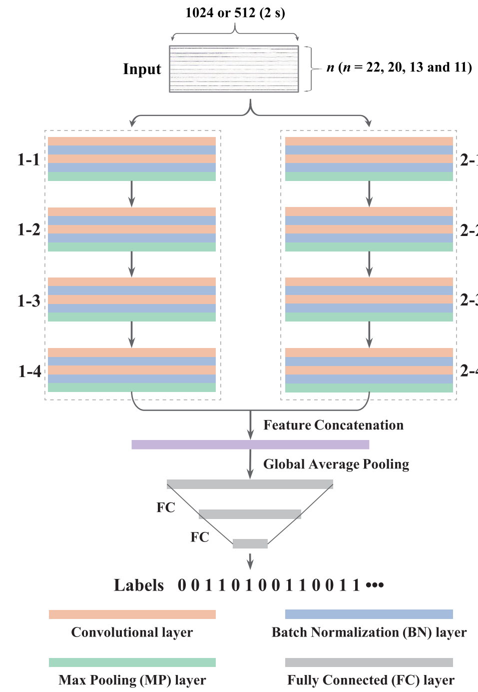

Fig. 4. Architecture of 1DCNN model with two parallel convolutional blocks and two FC layers for seizure detection.

During model training, the dropout rate of first FC layer is set to 0.25. The maximum number of epochs is 100, and the Early-Stopping (monitor = "val-loss" and patience = 10) is also used to prevent overfitting during the phase of model training. The model training and testing processing is implemented on an NVIDIA RTX4090 GPU with 24GB memory.

D. Model Training and Testing

K-fold cross validation (K-CV) combined with random selection and data augmentation (RS-DA) approach is performed during model training and testing (see Fig. 5). In the patient-specific strategy, K (2 to 7) is the number of seizures in each patient. For the ictal training samples from K-1 seizures, the data augmentation is carried out (as described in subsection III.B). For the interictal samples, they are first evenly divided into K parts. Among them, K-1 parts are used for training. Before model training, a certain amount of interictal samples are randomly selected from the K-1 parts to make the number equal to that of the ictal training samples. This process is called RS-DA (see Fig. 5). Then, one interictal part and one seizure are used for testing. After K folds, all interictal and ictal samples of a patient are tested.

In the patient-cross strategy, K is the number of the FLE patients or the TLE patients. To clarify the model training and testing process, we take the FLE patients as an example. For a selected FLE patient in the training set, DA is applied to augment the ictal samples (as detailed in subsection III-B). For the same FLE patient, RS is then used to select a certain amount of interictal samples, ensuring that their number is equal to that of the ictal training samples. This RS-DA procedure is repeated for K-1 FLE patients in the training set, while the remaining one FLE patient is reserved

TABLE III
PARAMETERS OF 1DCNN MODEL WITH TWO PARALLEL CONVOLUTIONAL BLOCKS AND TWO FC LAYERS

Type	Layers	Parameters	Type	Layers	Parameters
	Input	Form 1: 1000 (or 512) points*22 channels (whole-brain channels) Form 2: 1000 (or 512) points*20 channels (whole-brain channels excluding A1 and A2) Form 3: 1000 (or 512) points*13 channels (frontal-temporal channels with A1 and A2) Form 4: 1000 (or 512) points*11 channels (frontal-temporal channels only)			
1-1	Convolutional Convolutional MP	64 kernels, $s_1 = n\times 3$, $s_2 = 2$ 64 kernels, $s_1 = 3$, $s_2 = 2$ $s_1 = 2$, $s_2 = 2$	2-1	Convolutional Convolutional MP	64 kernels, $s_1 = n\times 5$, $s_2 = 2$ 64 kernels, $s_1 = 5$, $s_2 = 2$ $s_1 = 2$, $s_2 = 2$
1-2	Convolutional Convolutional MP	128 kernels, $s_1 = 3$, $s_2 = 1$ 128 kernels, $s_1 = 3$, $s_2 = 1$ $s_1 = 2$, $s_2 = 2$	2-2	Convolutional Convolutional MP	128 kernels, $s_1 = 5$, $s_2 = 1$ 128 kernels, $s_1 = 5$, $s_2 = 1$ $s_1 = 2$, $s_2 = 2$
1-3	Convolutional Convolutional MP	256 kernels, $s_1 = 3$, $s_2 = 1$ 256 kernels, $s_1 = 3$, $s_2 = 1$ $s_1 = 2$, $s_2 = 2$	2-3	Convolutional Convolutional MP	256 kernels, $s_1 = 5$, $s_2 = 1$ 256 kernels, $s_1 = 5$, $s_2 = 1$ $s_1 = 2$, $s_2 = 2$
1-4	Convolutional Convolutional MP	512 kernels, $s_1 = 3$, $s_2 = 1$ 512 kernels, $s_1 = 3$, $s_2 = 1$ $s_1 = 2$, $s_2 = 2$	2-4	Convolutional Convolutional MP	512 kernels, $s_1 = 5$, $s_2 = 1$ 512 kernels, $s_1 = 5$, $s_2 = 1$ $s_1 = 2$, $s_2 = 2$
	FC FC Output	256 neurons 64 neurons 2 neurons (0 for interictal samples and 1 for ictal samples)			

Note: The parameters for BN layers are set to default and not specified. Abbr: s_1, kernel size or pooling size; s_2, stride.

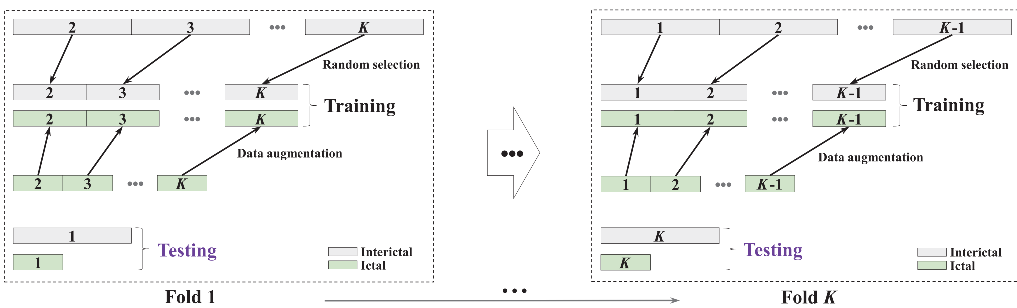

Fig. 5. K-CV combined with RS-DA for model training and testing in patient-specific and patient-cross strategies.

for testing (see Fig. 5). After K folds, all FLE patients are tested. For the TLE patients, the model training and testing procedures are the same as those applied to the FLE patients.

E. Method Evaluation

Two levels, event- and segment-based levels, are utilized simultaneously to comprehensively evaluate our method.

1) Event-Based Level: Three metrics, including sensitivity (Sen_e), false detection rate (FDR) and latency, are computed. The formulas for Sen_e and FDR are as follows:

$$\mathrm{Sen_e} = \frac{\text{Number of true detections}}{\text{Number of seizures}},$$

$$\mathrm{FDR} = \frac{\text{Number of false detections}}{\text{Hours of interictal EEG signals}}.$$

Latency is the time interval from the onset of a seizure to its detection (see Fig. 6). To give a reliable alarm, a simple postprocessing is therefore employed. The condition for triggering an alarm is having at least n positive labels within N consecutive detection labels. In this study, after comprehensive comparison and consideration, $n = 3$ and $N = 4$ are set for the patient-specific strategy, while $n = 4$ and $N = 5$ are set for the patient-cross strategy. Furthermore, to avoid unnecessary repeated alarms within a short period, we further optimize the postprocessing procedure by setting a minimum time interval

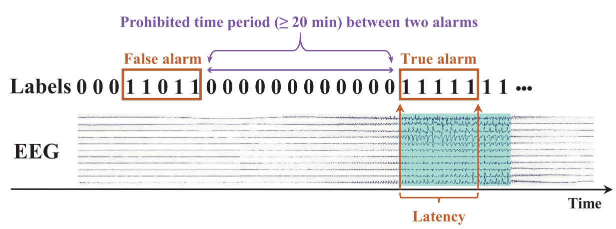

Fig. 6. Illustration for giving an alarm at the event-based level, and the minimum time interval between two alarms.

of 20 minutes between two detection alarms. This means that once the first alarm goes off, the second alarm must occur at least 20 minutes later (see Fig. 6).

2) Segment-Based Level: Four metrics, including sensitivity (Sen_s), specificity (Spec), accuracy (Acc) and area under curve (AUC), are calculated. Four metrics are computed based on interictal (0-negative) and ictal (1-positive) samples.

IV. RESULTS

After K-CV, the averaged classification results at two levels (event- and segment-based levels) are calculated for each patient. Since four different input forms (see Table III) are used, the optimal results can be selected from these four conditions for each patient. Therefore, the optimal results for each patient are summarized in this section.

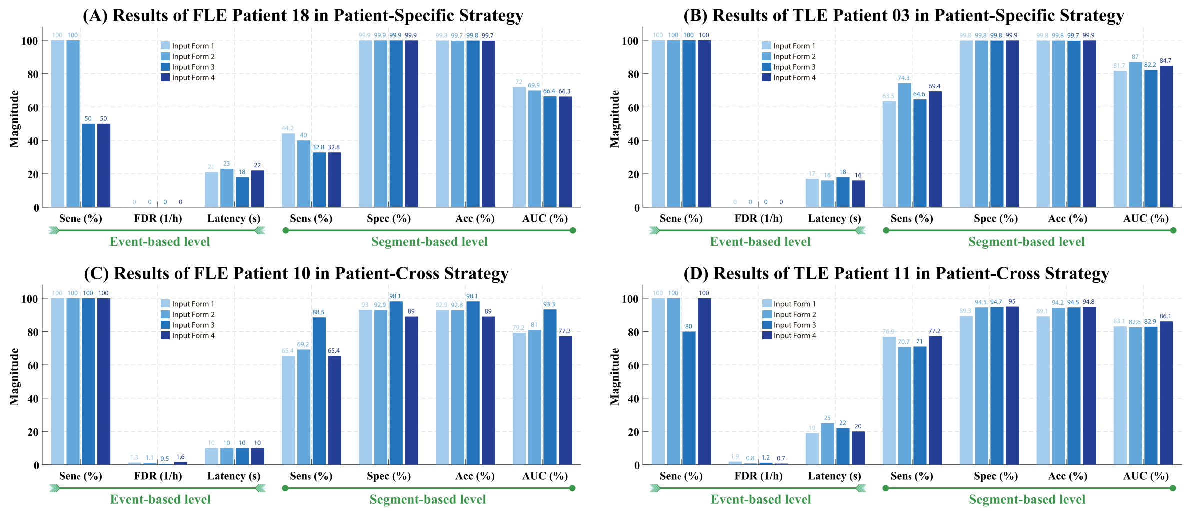

Fig. 7. Example of the optimal results selected for FLE and TLE patients in two strategies. (A) Optimal results finally selected from the input Form 1 for FLE patient 18 in patient-specific strategy; (B) Optimal results finally selected from the input Form 2 for TLE patient 03 in patient-specific strategy; (C) Optimal results finally selected from the input Form 3 for FLE patient 10 in patient-cross strategy; (D) Optimal results finally selected from the input Form 4 for TLE patient 11 in patient-cross strategy.

TABLE IV
OPTIMAL RESULTS FOR **FLE** IN PATIENT-**SPECIFIC** STRATEGY

				Event-based level			Segment-based level (%)			
No.	EEG (h)	#Sei	Form	Sen_e(%)	FDR(/h)	Lat(s)	Sen_s	Spec	Acc	AUC
01	8	2	3	100	0.0	20.0	49.8	99.7	99.5	74.7
02	16	2	4	100	0.0	16.0	52.5	99.7	99.7	76.1
03	8	4	3	100	0.0	12.5	56.1	99.4	99.2	77.7
04	8	5	4	100	0.0	13.2	72.0	98.0	97.8	85.0
05	8	3	3	100	0.0	18.7	63.7	99.7	99.6	81.7
06	4	6	2	100	0.0	11.0	75.6	99.6	99.3	87.6
07	16	2	2	100	0.0	14.0	36.4	99.6	99.5	68.0
08	8	2	4	50	0.0	12.0	34.4	99.4	99.3	66.9
09	14.3	2	4	100	0.0	24.0	46.0	99.5	99.4	72.7
10	8	2	4	100	0.0	8.0	82.1	99.6	99.6	90.9
11	16	2	4	100	0.0	11.0	89.7	99.4	99.3	94.5
12	8	3	4	66.7	0.0	18.0	34.0	99.7	99.6	66.8
13	16	2	2	100	0.0	16.0	78.5	99.7	99.7	89.1
14	8	2	2	50	0.0	8.0	30.8	99.6	99.5	65.2
15	8	2	4	0	0.0	–	12.5	99.2	99.2	55.9
16	16	2	2	100	0.0	19.0	58.5	99.1	99.0	78.8
17	16	2	1	100	0.0	14.0	83.6	99.7	99.6	91.7
18	8	2	1	100	0.0	21.0	44.2	99.9	99.8	72.0
19	8	2	4	100	0.0	10.0	85.3	99.8	99.7	92.6
20	4	2	3	100	0.0	16.0	74.7	99.8	99.6	87.3
All	206.3	51	–	90.2	0.0	14.9	58.0	99.5	99.4	78.8

TABLE V
OPTIMAL RESULTS FOR **TLE** IN PATIENT-**SPECIFIC** STRATEGY

				Event-based level			Segment-based level (%)			
No.	EEG (h)	#Sei	Form	Sen_e(%)	FDR(/h)	Lat(s)	Sen_s	Spec	Acc	AUC
01	16	2	3	100	0.0	16.0	82.1	99.9	99.8	91.0
02	12.9	2	2	100	0.0	12.0	70.4	99.7	99.6	85.0
03	16	2	2	100	0.0	16.0	74.3	99.8	99.8	87.0
04	11.3	2	4	100	0.0	21.0	85.3	99.9	99.8	92.6
05	10.1	2	1	100	0.0	37.0	55.5	98.9	98.6	77.2
06	16	2	1	100	0.0	9.0	94.4	99.9	99.9	97.2
07	8	4	4	100	0.0	18.0	36.4	98.4	97.8	67.4
08	6.1	4	2	100	0.0	10.5	85.6	99.5	99.4	92.6
09	8	2	1	100	0.0	21.0	27.5	99.6	99.4	63.6
10	8	3	3	100	0.0	16.0	85.3	99.8	99.7	92.5
11	8	5	4	100	0.0	10.4	87.7	99.7	99.6	93.7
12	8	3	3	100	0.0	12.7	90.4	98.8	98.7	94.6
13	8	2	4	100	0.0	18.0	41.7	99.9	99.6	70.8
14	8	7	2	100	0.0	12.9	73.3	99.3	99.0	86.3
15	15.3	2	2	100	0.0	10.0	79.8	99.9	99.9	89.8
16	16	2	3	100	0.0	17.0	80.9	99.3	99.2	90.1
17	16	2	2	100	0.0	9.0	64.6	99.8	99.7	82.2
18	16	2	3	100	0.0	20.0	84.9	99.7	99.7	92.3
19	11.9	2	4	100	0.0	17.0	76.2	99.9	99.8	88.0
20	16	2	3	100	0.0	25.0	28.8	99.9	99.8	64.3
All	235.6	54	–	100	0.0	16.4	70.3	99.6	99.4	84.9

A. Results of Patient-Specific Strategy

1) Results for FLE Patients: To better explain the selection process of the optimal results in FLE patients, an example of the results from four input forms for FLE patient 18 is given in Fig. 7(A). As shown in Fig. 7(A), compared to the input Forms 2, 3 and 4, the input Form 1 achieves a better overall performance at both event- (high sensitivity, same FDR and short latency) and segment-based (higher sensitivity, specificity, accuracy, and AUC) levels. Consequently, the results of input Form 1 are finally selected for FLE patient 18 and summarized in Table IV. As shown in Table IV, it summarizes the optimal results of 20 FLE patients, and at the event-based level, 46 out of 51 seizures (90.2% sensitivity) are correctly detected. The overall FDR and latency are 0.0/h and 14.9 s, respectively. At the segment-based level, the overall sensitivity of 58.0%, specificity of 99.5%, accuracy of 99.4%, and AUC of 0.788 are attained.

2) Results for TLE Patients: To better illustrate the process of selecting the optimal results in TLE patients, an example of the results from four input forms for TLE Patient 03 is showed in Fig. 7(B). As presented in Fig. 7(B), the superior overall performance is from the input Form 2 based on the results of both levels. Hence, the optimal results from the input Form 2 are selected for TLE patient 03 and summarized in Table V. Table V summarizes the optimal results of 20 TLE patients. As given in Table V, at the event-based level, 54 out of 54 seizures are truly detected, and the 100% sensitivity is therefore obtained. The 0.0/h FDR and the 16.4-sec latency are also achieved. At the segment-based level, we achieve an overall sensitivity, specificity, accuracy, and AUC of 70.3%, 99.6%, 99.4%, and 0.849, respectively. From Tables IV and V, we can see that the overall results of TLE patients are better than those of FLE patients at both levels.

TABLE VI
OPTIMAL RESULTS FOR **FLE** IN PATIENT-**CROSS** STRATEGY

				Event-based level			Segment-based level (%)			
No.	EEG (h)	#Sei	Form	Sen_e(%)	FDR(/h)	Lat(s)	Sen_s	Spec	Acc	AUC
01	8	2	2	100	2.0	29.0	55.7	89.5	89.4	72.6
02	16	2	3	100	1.0	14.0	72.2	91.3	91.3	81.8
03	8	4	4	100	1.6	13.5	69.0	87.9	87.9	78.5
04	8	5	1	100	2.7	12.4	63.2	71.6	71.5	67.4
05	8	3	2	100	1.1	16.0	77.3	96.8	96.7	87.1
07	16	2	4	50	1.9	18.0	34.1	92.7	92.6	63.4
08	8	2	2	50	1.3	10.0	53.6	96.0	96.0	74.8
09	14.3	2	1	100	1.5	16.0	51.0	87.6	87.5	69.3
10	8	2	3	100	0.5	10.0	88.5	98.1	98.1	93.3
12	8	3	3	66.7	2.3	22.0	58.1	81.2	81.2	69.7
13	16	2	1	100	2.4	19.0	78.8	79.5	79.5	79.2
14	8	2	3	100	1.5	12.0	78.0	94.9	94.8	86.4
15	8	2	2	0	2.9	–	77.3	61.0	61.0	69.2
17	16	2	3	100	1.2	21.0	90.9	94.9	94.8	92.9
18	8	2	3	100	1.2	22.0	33.3	91.5	91.4	62.4
19	8	2	4	100	1.8	10.0	92.6	87.3	87.3	89.9
20	4	2	1	100	0.6	23.0	64.1	98.7	98.5	81.4
All	170.3	41	–	87.8	1.6	16.7	66.9	88.3	88.2	77.6

TABLE VII
OPTIMAL RESULTS FOR **TLE** IN PATIENT-**CROSS** STRATEGY

				Event-based level			Segment-based level (%)			
No.	EEG (h)	#Sei	Form	Sen_e(%)	FDR(/h)	Lat(s)	Sen_s	Spec	Acc	AUC
01	16	2	4	100	1.0	16.0	88.9	94.4	94.4	91.7
03	16	2	1	100	1.0	13.0	90.5	95.5	95.5	93.0
05	10.1	2	4	100	1.2	50.0	66.5	95.6	95.4	81.1
06	16	2	1	100	0.3	10.0	100	96.8	96.8	98.4
07	8	4	1	75	0.8	16.7	49.1	95.2	94.9	72.2
08	6.1	4	1	100	1.1	14.5	87.9	93.4	93.4	90.7
09	8	2	4	100	0.3	29.0	57.3	96.2	96.1	76.8
10	8	3	4	100	1.0	16.0	89.2	93.7	93.7	91.4
11	8	5	4	100	0.7	20.0	77.2	95.0	94.8	86.1
12	8	3	3	100	0.5	22.0	86.1	97.5	97.3	91.8
13	8	2	4	100	1.6	10.0	88.5	94.0	93.9	91.2
14	8	7	1	100	1.4	13.7	82.3	89.0	88.9	85.6
15	15.3	2	1	100	0.2	12.0	78.0	98.4	98.3	88.2
16	16	2	3	100	0.7	29.0	75.9	95.6	95.6	85.8
17	16	2	1	100	0.3	13.0	87.5	97.1	97.1	92.3
18	16	2	1	100	0.3	21.0	86.0	99.0	98.9	92.5
19	11.9	2	2	100	0.7	17.0	90.3	97.0	97.0	93.7
20	16	2	2	100	1.7	14.0	68.2	89.8	89.8	79.0
All	211.4	50	–	98.0	0.8	18.8	80.5	95.2	95.1	87.9

B. Results of Patient-Cross Strategy

1) Results for FLE Patients: Since the sampling rate of FLE patients 6, 11 and 16 is 256 Hz (see Table I), the three FLE patients are removed. A total of 17 FLE patients are finally used for analysis. For better illustrating the selection of the optimal results in FLE patients, a typical instance of the results from four input forms for FLE patient 10 is given in Fig. 7(C). As seen in Fig. 7(C), we can see that the optimal results come from the input Form 3 for FLE patient 10 after comparing the results of four input forms. Therefore, for FLE patient 10, the results of the input Form 3 are finally selected and summarized in Table VI. As shown in Table VI, the optimal results of 17 FLE patients are given. At the event-based level, 36 out of 41 seizures (87.8% sensitivity) are finally detected, and 1.6/h FDR and 16.7-sec latency are attained. At the segment-based level, we attain an overall sensitivity, specificity, accuracy, and AUC of 66.9%, 88.3%, 88.2% and 0.776, respectively.

2) Results for TLE Patients: TLE patients 2 and 4 are removed due to the sampling rate of 256 Hz (see Table II). Consequently, 18 TLE patients are finally utilized for analysis. In TLE patients, we also provide an example for selecting the optimal results (see Fig. 7(D)). As illustrated in Fig. 7(D), the optimal results for TLE patient 11 are from the input Form 4 after comparing the performances of four input forms at both levels, and the corresponding results are therefore summarized in Table VII. As demonstrated in Table VII, the optimal results of 18 TLE patients are given. At the event-based level, 49 out of 50 seizures are finally detected, and a sensitivity of 98.0% is obtained. The FDR and the latency are 0.8/h and 18.8 s, respectively. At the segment-based level, we attain 80.5% sensitivity, 95.2% specificity, 95.1% accuracy and 0.879 AUC. From the results in Tables VI and VII, the overall performances of TLE patients are also better than those of FLE patients.

V. DISCUSSION

In this section, we first compare the performances of four input forms in patient-specific strategy, and then compare the performances of four input forms in patient-cross strategy. Finally, we compare our work with prior studies and briefly discuss the limitations of our work.

A. Comparison of Four Input Forms in Patient-Specific Strategy

Table VIII summarizes the overall results of four input forms for FLE group. At the event-based level, the input Forms 1 and 3 achieve the highest sensitivity of 82.4%, while the input Form 2 records the lowest sensitivity at 78.4% but the shortest latency of 16.6 s. At the segment-based level, the input Form 4 demonstrates superior overall performance, attaining 54.4% sensitivity, 99.5% specificity, 99.4% accuracy, and an AUC of 0.769.

Table VIII also presents the overall results for TLE group. At the event-based level, the input Forms 1, 2, and 4 achieve the highest sensitivity of 96.3%, whereas the input Form 3 has the lowest sensitivity of 94.4% and the longest latency of 18.4 s. At the segment-based level, the input Form 2 exhibits the best overall performance, achieving 67.9% sensitivity, 99.6% specificity, 99.4% accuracy, and an AUC of 0.838. Moreover, it should be noted that the overall results across four input forms are relatively similar, with no significant differences.

Under the patient-specific strategy, we can observe that across four different input forms, the detection performance for TLE seizures is better than that for FLE seizures. For such a finding, one major reason we suspect is that FLE seizures generally have shorter durations (see Tables I and II), which makes it more challenging to detect shorter FLE seizures, leading to lower sensitivity. Another possible reason is whether a more complex DL network needs to be constructed to improve the detection performance of FLE seizures.

B. Comparison of Four Input Forms in Patient-Cross Strategy

Based on the patient-cross strategy, Table IX summarizes the overall results of four input forms for FLE and TLE groups.

TABLE VIII
OVERALL RESULTS OF FOUR INPUT FORMS IN PATIENT-SPECIFIC STRATEGY FOR FLE AND TLE GROUPS

Form	Event-based level			Segment-based level (%)			
	Sen_e (%)	FDR (/h)	Lat (s)	Sen_s	Spec	Acc	AUC
FLE Group							
1	82.4 (42/51)	0.0	17.6	49.6	99.6	99.5	74.6
2	78.4 (40/51)	0.0	16.6	51.4	99.5	99.4	75.5
3	82.4	0.0	17.4	52.3	99.5	99.4	75.9
4	80.4 (41/51)	0.0	17.2	54.4	99.5	99.4	76.9
Optimal	90.2 (46/51)	0.0	14.9	58.0	99.5	99.4	78.8
TLE Group							
1	96.3 (52/54)	0.0	17.3	65.7	99.6	99.5	82.7
2	96.3	0.0	17.7	67.9	99.6	99.4	83.8
3	94.4 (51/54)	0.0	18.4	66.9	99.6	99.4	83.2
4	96.3	0.0	18.0	66.3	99.6	99.5	83.0
Optimal	100 (54/54)	0.0	16.4	70.3	99.6	99.4	84.9

TABLE IX
OVERALL RESULTS OF FOUR INPUT FORMS IN PATIENT-CROSS STRATEGY FOR FLE AND TLE GROUPS

Form	Event-based level			Segment-based level (%)			
	Sen_e (%)	FDR (/h)	Lat (s)	Sen_s	Spec	Acc	AUC
FLE Group							
1	78.1 (32/41)	1.6	16.5	60.4	87.6	87.5	74.0
2	85.4 (35/41)	1.7	16.2	64.8	85.9	85.9	75.3
3	82.9 (34/41)	1.6	18.0	62.4	86.7	86.6	74.5
4	80.5 (33/41)	1.7	16.6	63.8	84.5	84.5	74.2
Optimal	87.8 (36/41)	1.6	16.7	66.9	88.3	88.2	77.6
TLE Group							
1	96.0 (48/50)	0.9	18.6	78.5	94.8	94.8	86.6
2	96.0	1.0	20.1	78.2	93.9	93.8	86.1
3	94.0 (47/50)	1.1	18.4	79.3	93.3	93.2	86.3
4	96.0	1.0	21.0	78.6	93.7	93.6	86.2
Optimal	98.0 (49/50)	0.8	18.8	80.5	95.2	95.1	87.9

In FLE group, the input Form 2 demonstrates superior performance at the event-based level, achieving 85.4% sensitivity, an FDR of 1.7/h, and the shortest latency of 16.2 s. At the segment-based level, the input Form 2 also attains the highest sensitivity (64.8%) and the largest AUC (0.753), while the input Form 3 achieves the highest specificity (87.6%) and accuracy (86.6%).

In TLE group, at the event-based level, the input Forms 1, 2, and 4 reach the highest sensitivity of 96.0%, albeit with longer latencies of 18.6-21 s, whereas the input Form 3 records the lowest sensitivity (94.0%) but the shortest latency (18.4 s). At the segment-based level, the input Form 3 achieves the highest sensitivity (79.3%), but the input Form 1 outperforms in specificity (94.8%), accuracy (86.6%) and AUC (0.866). Notably, the overall results across four input forms remain comparable, with differences of no more than 1.6% at the segment-based level.

Under the patient-cross strategy, we can also observe that the detection sensitivities for TLE seizures are generally better than those for FLE seizures across four input forms. For the phenomenon of patient-cross strategy appearing similar to the patient-specific strategy, we believe that the two possible reasons are the same as those discussed under the patient-specific strategy condition and, therefore, will not be elaborated on again here.

C. Compared to Studies Focusing on Detection of FLE or TLE Seizures

Table X summarizes the studies focusing on the detection of FLE or TLE seizures. As summarized in Table X, the studies [46], [47], [48] detail the seizure detection outcomes for only TLE patients, and the studies [49], [50], [51] detail the seizure detection outcomes for both FLE and TLE patients. For FLE patients, the studies [49], [50], [51] achieve 26.9-60.87% sensitivity and 0.17-1.09/h FDR based on the patient-cross strategy. For TLE patients, the studies [47], [48], [49], [50], [51] attain 77.4-96.0% sensitivity and 0.14-1.07/h FDR based on the patient-cross strategy. Compared to the above results, wen can see that our method achieve remarkable performances with a higher sensitivity of 87.8% but a higher 1.6/h FDR for FLE patients, and a higher sensitivity of 98.0% with a low 0.8/h FDR for TLE patients based on the patient-cross strategy. Additionally, an interesting finding is that the studies [49], [50], [51], as well as our work, show better performances in detecting TLE seizures compared to FLE seizures. It is also necessary to emphasize that our work is the first to use 1DCNN methods for a comprehensive analysis of FLE and TLE seizure detection, while the studies [46], [47], [48], [49], [50], [51] focus on using conventional ML or threshold-based methods.

Although our work shows remarkable performances in seizure detection of FLE and TLE, there are several limitations that should be emphasized for future research directions. One limitation is that the current work only uses one 1DCNN model for seizure detection. Other DL models, such as LSTM, CNN-LSTM, etc., can also be used for further comparison. The second limitation is that the EEG data volume of FLE and TLE is not large enough (only 20 FLE and 20 TLE patients). Therefore, more FLE and TLE EEG data can be collected and further analyzed in conjunction with our proposed method. Additionally, these newly collected EEG data can be combined with other DL models for further analysis and comprehensive comparison.

D. Compared to Studies Using 1DCNN-Related Methods for Seizure Detection

Table XI summarizes the state-of-the-art research employing 1DCNN-related methods for seizure detection. As given in Table XI, the studies [29], [30], [31], [32], [44] report sensitivity ranging from 95% to 97.80%, specificity between 90% and 99.81%, and accuracy from 82.00% to 99.96%. Although the above results are commendable, these studies are based on a small-sample Bonn dataset (comprising 500 single-channel EEG segments, each lasting 23.6 s, including 100 seizure segments). In real-world clinical practice, EEG data from people with epilepsy is typically collected through long-term continuous monitoring. Therefore, seizure detection based on long-term continuous EEG data holds greater practical significance. The study [28], based on the patient-specific strategy, reports 88.14% sensitivity, 99.62% specificity, and 99.54% accuracy for the CHI-MIT scalp EEG dataset, and 90.09% sensitivity, 99.81% specificity, and 99.73% accuracy for the SWEC-ETHZ iEEG dataset. Due to the inherent limitations in the datasets (e.g., unspecified seizure types per individual, varying numbers of channels, etc.), this study omits further analysis under the patient-cross strategy.

Unlike the above studies using 1DCNN-related methods for general seizure detection, this work focuses on classifying specific seizures–namely, FLE and TLE seizures. By also

TABLE X
COMPARISON OF PRIOR STUDIES AND THIS WORK ON DETECTION OF FLE OR TLE SEIZURES AT EVENT-BASED LEVEL

Authors	Feature	Classifier	Strategy*	#Patients	EEG (h)	#Seizures	Sen_e (%)	FDR (/h)	Latency (s)
van Putten et al. (2005) [46]	Time/Frequency domain features	PRTools	1	16 TLE	40	–	33.0-99.0	1	–
Hocepied et al. (2013) [47]	Nonstationarity	Threshold	2	5 TLE	28.58	30	96.0	0.14	14.3
Fürbass et al. (2017) [48]	Multimodal features	EpiScan+Threshold	2	55 TLE	–	284	94.0	1.07	–
Fan et al. (2019) [49]	Parameter identification of an NMM	Threshold	2	8 FLE	115.53	23	60.87	0.17	-3.25
			2	12 TLE	237.09	60	93.33	0.28	38.8
Koren et al. (2021) [50]	Raw EEG	Three software packages	2	14 FLE	1259	69	50.7	1.09	–
			2	46 TLE	3893	419	86.4	0.9	–
Elezi et al. (2022) [51]	Raw EEG	EpiScan	2	10 FLE	–	26	26.9	–	–
			2	37 TLE	–	115	77.4	–	–
This work	2-sec time-channel EEG segments	1DCNN	1	20 FLE	206.3	51	90.2	0.0	14.9
			1	20 TLE	235.6	54	100	0.0	16.4
			2	17 FLE	170.3	41	87.8	1.6	16.7
			2	18 TLE	211.4	50	98.0	0.8	18.8

*1 for patient-specific strategy and 2 for patient-cross strategy. NMM, Neural Mass Model.

TABLE XI
COMPARISON OF PRIOR STUDIES AND THIS WORK USING 1DCNN-RELATED METHODS FOR SEIZURE DETECTION AT SEGMENT-BASED LEVEL

Authors	Feature	Classifier	Dataset	Strategy*	#Patients	EEG (h)	#Seizures	Sen_s (%)	Spec (%)	Acc (%)
Acharya et al. (2018) [29]	23.6-sec single-channel EEG	Deep 1DCNN	Bonn	–	–	3.28	–	95	90	88.7
Ullah et al. (2018) [31]	3-sec single-channel EEG	P-1DCNN	Bonn	–	–	3.28	–	–	–	99.1
Xu et al. (2020) [44]	23.6-sec single-channel EEG	1DCNN-LSTM	Bonn	–	–	3.28	–	–	–	82.00-99.39
Wang et al. (2021) [32]	1-sec single-channel EEG	1DCNN	Bonn	–	–	3.28	–	97.80	99.81	99.41
Wang et al. (2021) [28]	2-sec time-channel EEG	1DCNN	CHB-MIT	1	24	518.6	145	88.14	99.62	99.54
			SWEC-ETHZ	1	18	432	126	90.09	99.81	99.73
Wang et al. (2023) [30]	1-sec single-channel EEG	MFO-1DCNN	Bonn	–	–	3.28	–	–	–	86.16-99.96
This work	2-sec time-channel EEG	1DCNN	Private	1	20 FLE	206.3	51	58.0	99.5	99.4
			Private	1	20 TLE	235.6	54	70.3	99.6	99.4
			Private	2	17 FLE	170.3	41	66.9	88.3	88.2
			Private	2	18 TLE	211.4	50	80.5	95.2	95.1

Abbreviation explanation: P-1DCNN, Pyramidal 1DCNN; MFO-1DCNN, Moth-flame optimization 1DCNN.

employing 1DCNN approaches, our study evaluates performances under both patient-specific and patient-cross strategies during seizure detection. Under the patient-specific strategy, our work achieves the same high accuracy of 99.4% for both the FLE and TLE groups. Moreover, under the patient-cross strategy, it also attains high accuracies of 88.2% and 95.1% for the FLE and TLE groups, respectively. Compared to the results from the general seizure detection studies [28], [29], [30], [31], [32], [44] (see Table XI), the results of both strategies in this work demonstrate that 1DCNN-based methods can effectively detect specific types of epileptic seizures, such as FLE and TLE seizures. This further indicates that exploring detection methods for specific types of epileptic seizures is both feasible and effective, revealing the potential to advance tailored monitoring and enable more precise interventions in epilepsy diagnosis. In addition, it is also important to emphasize that, to the best of our knowledge, this study is the first to apply 1DCNN methods to the detection of both FLE and TLE seizures simultaneously.

VI. CONCLUSION

In this paper, we studied seizure detection using scalp EEG signals from 20 FLE patients and 20 TLE patients, and a 1DCNN model with two parallel convolutional blocks was built for classification. Furthermore, our work discussed and compared the seizure detection results of FLE and TLE in patient-specific strategy and patient-cross strategy, respectively. From the results of each strategy, the overall performances of TLE were better than those of FLE at both evaluation levels. Moreover, through the comparison of four input forms, the optimal input form can be selected for each individual, effectively improving the final seizure detection results and outperforming any single input form. This highlights the importance of channel selection in seizure detection analysis. All of these may provide a reference for the seizure detection of FLE and TLE in future research.

CODE AVAILABILITY

The codes of both strategies for FLE and TLE seizure detection analysis can be obtained from here: *FLE and TLE Seizure Detection Using 1DCNN*.

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